

概率统计 21-22-3(A)标准答案及评分标准

一、选择题

1)C 2) B 3)D 4) B, 5) B

二、填空题

1)9/16=0.5625;

2)27/64=0.422

3) 14

4)0.0228

5)-1/6

6) $-0.4\sqrt{2}$

7) 0.3

8) $7/8=0.875$

$$9) F(x) = \begin{cases} 0 & x < -2 \\ 0.3 & -2 \leq x < -1 \\ 0.6 & -1 \leq x < 1 \\ 1 & x \geq 1 \end{cases}$$

$$10) f_Y(y) = \begin{cases} \frac{1}{2} \sin\left(\frac{1-y}{2}\right) & 1-\pi < y < 1 \\ 0 & \text{其它} \end{cases}$$

11) $t(1)$

12) [149.175, 150.825]

13) $\frac{1}{3} = 0.33$

三、

$$(1) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1; \dots\dots\dots$$

$$a \int_{-1}^0 \int_{-\sqrt{-y}}^{\sqrt{-y}} dy dx = 1; \dots\dots\dots$$

$$a = \frac{3}{4} \dots\dots\dots$$

$$(2) f_X(x) = \int_{-\infty}^{\infty} f(x, y) dy \dots\dots\dots$$

$$\text{当 } 0 < |x| < 1 \text{ 时 } f_X(x) = \int_{-1}^{-x^2} a dy = \frac{3}{4}(1 - x^2) \dots\dots\dots$$

$$\text{当 } |x| \geq 1 \text{ 时 } f_X(x) = 0. \dots\dots\dots$$

$$(3) f_{Y|X}(y|x) = \frac{f(x, y)}{f_X(x)} = \begin{cases} \frac{1}{1-x^2} & -1 < y < -x^2 \\ 0 & \text{其它} \end{cases} (|x| < 1)$$

$$f_{Y|X}(y|0.5) = \begin{cases} \frac{4}{3} & -1 < y < -0.25 \\ 0 & \text{其它} \end{cases} \dots\dots\dots$$

$$P(-0.75 < Y < -0.15 | X = 0.5) = \int_{-0.75}^{-0.15} f_{Y|X}(y|0.5) dy = \int_{-0.75}^{-0.25} \frac{4}{3} dy = \frac{2}{3}$$

四、 A_i 表示事件：加入的两个球中有*i*个白球； B 表示事件：抽到白球. 则

$$P(A_i) = \frac{1}{3}; i = 0, 1, 2$$

$$P(B|A_i) = \frac{2+i}{7};$$

$$(1) P(B) = P(A_0)P(B|A_0) + P(A_1)P(B|A_1) + P(A_2)P(B|A_2)$$

$$= \frac{1}{3} \left(\frac{2}{7} + \frac{3}{7} + \frac{4}{7} \right) = \frac{3}{7}$$

(2)

$$P(A_2|B) = \frac{P(A_2)P(B|A_2)}{P(B)}$$

$$= \frac{\frac{1}{3} \times \frac{4}{7}}{\frac{3}{7}} = \frac{4}{9} \approx 0.444$$

五、 Z 的分布函数 $F_Z(z) = P(Z \leq z) = P(X + 2Y \leq z)$

$$= \iint_{x+2y \leq z} f(x, y) dx dy$$

当 $z < 0$ 时, $F_Z(z) = 0$;

当 $0 \leq z \leq 2$ 时, $F_Z(z) = \iint_{x+2y \leq z} f(x, y) dx dy$

$$= 0.5 \int_0^{\frac{z}{2}} \int_0^{z-2y} dx dy$$

$$= \frac{z^2}{8}$$

当 $2 < z \leq 4$ 时

$$F_Z(z) = \iint_{x+2y \leq z} f(x, y) dx dy = 1 - P(X + 2Y > z)$$

$$= 1 - \frac{1}{8}(4-z)^2$$

当 $z > 4$ 时, $F_Z(z) = 1$

Z 的概率密度为

$$f_Z(z) = [F_Z(z)]' = \begin{cases} \frac{z}{4} & 0 < z \leq 2 \\ \frac{1}{4}(4-z) & 2 < z < 4 \\ 0 & \text{其它} \end{cases}$$

六、 X_i 表示第*i* 页错别字的个数, $i = 1, 2, \dots, 100$.
 $X_i \sim P(0.2)$; $\mu = EX_i = 0.2$; $\sigma^2 = DX_i = 0.2$; $n = 100$
 所求概率为:

$$P\left(\sum_{i=1}^n X_i \leq 25\right) \approx \Phi\left(\frac{25 - n\mu}{\sqrt{n}\sigma}\right)$$

$$= \Phi\left(\frac{5}{\sqrt{20}}\right) = \Phi\left(\frac{\sqrt{5}}{2}\right)$$

七、记 $f_X(x, \mu) = \frac{1}{2\sqrt{\pi}} e^{-\frac{(x-\mu)^2}{4}}$; $f_Y(y, \mu) = \frac{1}{2\sqrt{2\pi}} e^{-\frac{(y-2\mu)^2}{8}}$

(1) 似然函数为: $L(\mu) = \prod_{i=1}^m f_X(X_i, \mu) \prod_{j=1}^n f_Y(Y_j, \mu)$

$$= C e^{-\sum_{i=1}^m \frac{(X_i - \mu)^2}{4} - \sum_{j=1}^n \frac{(Y_j - 2\mu)^2}{8}}$$

其中C是与参数 μ 无关的常数。

对数似然: $\ln L(\mu) = \ln C - \sum_{i=1}^m \frac{(X_i - \mu)^2}{4} - \sum_{j=1}^n \frac{(Y_j - 2\mu)^2}{8}$

对对数似然求导数并令导数等于 0:

$$l(\mu)' = \frac{1}{2} \sum_{i=1}^m (X_i - \mu) - \frac{1}{2} \sum_{j=1}^n (Y_j - 2\mu) = 0$$

解得: $\hat{\mu} = \frac{m\bar{X} + n\bar{Y}}{m + 2n}$; 其中 $\bar{X} = \frac{1}{m} \sum_{i=1}^m X_i$; $\bar{Y} = \frac{1}{n} \sum_{j=1}^n Y_j$.

(2) $E\hat{\mu} = E \frac{m\bar{X} + n\bar{Y}}{m + 2n} = \frac{mE\bar{X} + nE\bar{Y}}{m + 2n}$

$$E\bar{X} = \mu; E\bar{Y} = 2\mu;$$

所以, $E\hat{\mu} = \frac{m\mu + n \times 2\mu}{m + 2n} = \mu$.

故, $\hat{\mu}$ 是 μ 的无偏估计量。

八、(1) $n = 25, \alpha = 0.05$,

检验统计量 $T = \frac{\bar{X} - 48}{S_n} \sqrt{n} | H_0 \sim t(n-1)$

拒绝域: $D = \{|T| > t_{\frac{\alpha}{2}}(n-1)\} = \{|T| > 2.064\}$

$\bar{x} = 50, s_n = 3$

T 的观测值: $T = \frac{50-48}{3} \sqrt{25} = 3.33$

$$|3.33| > 2.064$$

所以, 拒绝原假设。

(2) σ^2 的置信度为 95% 的置信区间为: $\left[\frac{(n-1)S_n^2}{\chi_{0.025}^2(24)}, \frac{(n-1)S_n^2}{\chi_{0.975}^2(24)} \right]$

$$= \left[\frac{24 \times 9}{39.36}, \frac{24 \times 9}{12.4} \right] = [5.488, 17.419]$$